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Research paper

The EKF-based SINS/DVL integrated navigation for AUV on lie group under hovring condition☆

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ABSTRACT

When autonomous underwater vehicles (AUVs) perform tasks under hovering conditions, the strapdown inertial navigation system (SINS) and doppler velocity log (DVL) integrated navigation system encounters significant observability constraints. It is not feasible to enhance observability through motion excitation or the addition of external sensors, leading to partial unobservability of the system's state variables. This results in inconsistencies during the extended kalman filter (EKF) process, causing filter divergence. To address this issue, this paper proposes a SINS/DVL integrated navigation algorithm based on Lie group, redefining the system's nonlinear state errors and measurement models. To circumvent the limitations of conventional inertial frames, a new auxiliary vector is introduced in the local-level frame to achieve affine improvement, and a lie group-based EKF based on group affine is obtained. This approach not only effectively reduces inconsistencies caused by unobservable state variables but also enhances the stability of the filtering system under hovering conditions. Finally, theoretical analysis and simulation results demonstrate that the proposed algorithm offers certain advantages in terms of heading attitude estimation and positioning accuracy.

1. Introduction

As land resources have gradually become depleted in recent years, the vast resources of the ocean have attracted increasing attention (Sunagawa et al., 2020)- (Jameian et al., 2019). The oceans contain rich deposits of minerals, marine energy, and abundant biological resources. However, the sheer vastness of the ocean presents significant challenges in advancing technologies for ocean exploration and development.

The exploration and development of the ocean are inextricably linked to the support of advanced platforms, with autonomous underwater vehicles (AUVs) playing a significant role (Lv et al., 2020)- (He et al., 2023). AUVs are utilized for various tasks such as resource exploitation, terrain mapping, and seabed equipment inspection in unknown and complex environments. The real-time capture of reliable navigation information is crucial for enhancing the accuracy of AUVs during underwater operations.

Due to the distinct characteristics of the underwater environment compared to land-based vehicles' locations, underwater navigation is confronted with the rapid attenuation of electromagnetic waves, rendering global satellite navigation systems ineffective. Consequently, there is a scarcity of viable sensors for underwater AUV navigation (González-García et al., 2020; Allotta et al., 2016). Acoustic-based long baseline (LBL) and ultra-short baseline (USBL) navigation systems require pre-deployment, which limits their applicability (Zhu et al., 2020). As a result, AUV typically employ a Strapdown Inertial Navigation System (SINS) as the primary navigation system. The SINS system compensates for its cumulative errors by integrating velocity information from a doppler velocity log (DVL) sensor, thereby achieving SINS/DVL integrated navigation (J Snyder, 2010)- (Fukuda et al., 2021).

The extended kalman filter (EKF) is a commonly employed filtering algorithm for SINS/DVL integrated navigation systems (Liu et al., 2021)- (Liu et al., 2014). The EKF typically utilizes a first-order Taylor

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approximation for linearization, neglecting the influence of higher-order terms to facilitate the integration process (Llerena Cana et al., 2021). However, the issue of inconsistency in EKF has received limited attention in research. Julier and Uhlmann were among the first scholars to identify the problem of scale inconsistency in EKF-based simultaneous localization and mapping (SLAM), which manifests as oscillatory divergence in estimation results (Julier and Uhlmann, 1997). Subsequent studies have demonstrated that this inconsistent estimation arises from errors introduced during the linearization process, which are inherently unavoidable (Bailey et al., 2006).

The aforementioned studies primarily focus on terrestrial SLAM applications, with limited research addressing the inconsistency issue of EKF in SINS/DVL integrated navigation systems. In fact, this problem is intrinsically related to the system's observability, as theoretical observability analysis must precede the filtering process to determine whether inconsistent estimation issues exist. Both (Lee et al., 2020) and (Delaune et al., 2021) have investigated visual-inertial odometry (VIO) systems under non-stationary motion conditions, proposing methods to enhance observability through coordinate system transformation and additional sensors, respectively. In (Itzik and Diamant, 2015), the observability Gramian method was utilized to analyze an integrated navigation system combining an Inertial Navigation System (INS), Doppler Velocity Log (DVL), and pressure sensor (PS). The study demonstrated that the incorporation of additional sensors or the implementation of motion excitation techniques, such as rotational maneuvers, significantly enhances system observability, even under stationary conditions. However, when AUVs perform oceanographic exploration or geological survey missions, they are often required to maintain prolonged stationary positioning and attitude control near target areas, necessitating stable hovering capabilities (Gao et al., 2024). These operational constraints preclude the possibility of motion excitation or surfacing for correction, while simultaneously limiting the availability of additional sensors (Wang et al., 2024). Consequently, under the motion-constrained hovering conditions of AUVs, there is a compelling need to develop novel integrated navigation methodologies that address the state estimation inconsistency inherent in traditional EKF approaches.

In recent years, the development of Lie group and manifold theories has led to the emergence of novel filtering techniques that extend the solvability of algebraic equations to differential equations (Jiwari et al., 2022). The invariant filter has achieved notable success in the SLAM domain, providing effective visual-inertial SLAM methods by leveraging system invariance (Brossard et al., 2018). Traditional EKF implementations usually establish the attitude errors in a special orthogonal group (SO(3)), are susceptible to bias in dependent state estimation matrices when initial state values (linearization points) contain significant errors, due to first-order linearization and discretization processes. Current research on Lie groups and Lie algebras predominantly addresses large misalignment angle problems (Chang et al., 2021a)-(Hashim and Eltoukhy, 2021), as the Lie group error satisfies " log-linear " differential equations, inherently immune to nonlinearities induced by substantial initial state errors. This approach eliminates the need to consider system observability conditions, since dynamic excitation exacerbates the impact of initial state errors (Sheng et al., 2024). However, limited research has been conducted on integrated navigation under low-observability conditions. Barrau has demonstrated that Lie group-based error definitions possess excellent error self-consistency, with state estimation independent of the state matrix (Barrau and Bonnabel, 2017). This characteristic has begun to find applications in navigation, though it is crucial to note that this property relies on the group state model satisfying the group affine theorem. In (Herrmann and Kotyczka, 2024), the authors developed theoretical models based on inertial frames by leveraging Lie group theory. Potokar et al. (2021) proposed an invariant extended kalman filter (InEKF) approach combined with inertial measurement units (IMUs) and DVL sensor fusion in the inertial frame, extending EKF applications through the log-linear differential equation property of errors, representing a cross-domain application. However, it is precisely because the inertial frame completely ignores the rotation of the earth and the coriolis force effect that the group affine equation can be established. For navigation-grade SINS systems, it is necessary to consider vehicle motion relative to Earth rather than inertial space, making coordinate system design crucial for maintaining group affine property. The group affine theorem is applied in the Earth-Centered Earth-Fixed (ECEF) frame but limited its application to pre-integration (Barrau and Bonnabel, 2020). Chang et al. also implemented INS mechanization modifications to satisfy group affine property in the earth frame (Chang et al., 2021b). This observation underscores the significance of investigating approaches to achieve group affine property across different frames, which warrants further systematic research and exploration.

In summary, this study addresses the engineering challenge of SINS/DVL integrated navigation for AUVs under hovering conditions, where conventional methods of enhancing observability through motion excitation or additional sensors are unfeasible. Based on theoretical observability analysis, we propose a novel Lie group and Lie algebra-based EKF for integrated navigation filtering, aiming to mitigate the impact of inconsistency on navigation performance. Specifically, the main contributions of this work are as follows.

(1) Distinct from the conventional observability method that can only obtain the quantity of unobservable states, this paper proposes an observability analysis method based on right-zero space analysis of piece-wise constant system (PWCS) combined with singular value decomposition (SVD) to obtain which specific unobservable states. And based on this, it reveals and validates the explanation of the inconsistency and the reason for the SO(3)-based EKF (SO-EKF) when the SINS/DVL system is in hovering condition. And it is emphasized that for the purpose of the study, which is different from other conventional literature, we do not consider how to improve the observable states. This is because it is not possible to perform motion excitation or add sensors to improve observability in this hovering condition task. The purpose of this paper is to minimize the negative impact of unobservable states on the integrated navigation system without any other external assistance.

(2) Aiming at the AUV hovering condition which has not received sufficient attention, a SINS/DVL integrated navigation method is proposed, which introduces the Lie group and Lie algebra theory. This paper doesn't derive the " log-linear " advantage of this method in high dynamics from the perspective of initial state error, but based on the condition that the carrier is constrained by motion under poor-observation. The state space is modeled as a matrix lie group of EKF, and redefines the nonlinear state error and measurement model. Moreover, different from the simplified calculation of the inertial frame, the influence of the earth's rotation on navigation is considered based on the local-level frame to face the demand of high-precision applications. The auxiliary velocity vector and gravity correction term are introduced. Not only the group affine property is realized, but also the theoretical accuracy of the complete system is further improved. This is the first attempt to propose a SINS/DVL integrated navigation method based on EKF for AUV hovring that satisfies lie group affine in a local-level frame.

The rest of this paper is organized as follows. Section 2 introduces the SINS mechanical arrangement and dynamic model in local-level frame, emphasizing the distinction from conventional inertial frame. Section 3 introduces the observability analysis based on SINS/DVL integrated navigation system,and proposes an observability analysis method based on right-zero space analysis of PWCS combined with SVD, which obtains the problem of inconsistency in the SO-EKF. Section 4 propose a SINS/DVL integrated navigation algorithm based on SE(3) of EKF with lie-group affine (LSEGA-EKF). Section 5 compares the performance of LSEGA-EKF with other methods through Monte Carlo simulations and field experiments, and the results are discussed. Finally, Section 6 outlines the main conclusions of this work and interesting directions for future research.

2. SINS mechanization and reference frames

Different frames determine different SINS mechanizations. The conventional inertial frame actually simplifies the formula. This section derives the SINS mechanizations and error state space model based on the local-level frame.

2.1. SINS mechanization in local-level frame

In this paper, the local-level frame is the navigation frame n, b represents the body frame, and i is the inertial frame. According to the similarity transformation (Li and Li, 2020), the attitude update equation is derived as:

$$\dot{\mathbf{C}}_b^n = \mathbf{C}_b^n[\boldsymbol{\omega}_{nb}^b \times] - [\boldsymbol{\omega}_{in}^n \times]\mathbf{C}_b^n \tag{1}$$

In the equation, $\boldsymbol{\omega}_{ib}^b$ represents the angular velocity of the carrier; $\omega_{in}^n = \omega_{ie}^n + \omega_{en}^n$ represents the angular velocity of the motion in the n frame, which is caused by the rotation of the earth and the linear motion of the body. The velocity differential equation can be obtained from the basic specific force equation:

$$\dot{\boldsymbol{v}}^n = \boldsymbol{C}_b^n \boldsymbol{f}^b - (2\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n) \times \boldsymbol{v}^n + \boldsymbol{g}^n \tag{2}$$

$\boldsymbol{f}^b$ is the specific force vector, and $\boldsymbol{g}_{ib}^n$ is the local gravity vector.

The position differential equation is

$$\dot{\boldsymbol{P}}^n = \begin{bmatrix} \dot{L} \\ \dot{\lambda} \\ \dot{h} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{R_M+h} & 0 \\ \frac{\sec L}{R_N+h} & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_E \\ \boldsymbol{v}_N \\ \boldsymbol{v}_U \end{bmatrix} \tag{3}$$

L, λ and h are latitude, longitude, and altitude respectively. The navigation parameters can be calculated in real time through the above update equations.

2.2. Error state model of SINS based on SO-EKF

According to the SINS update equation in the previous section, the corresponding error equation can be derived. The differential equation of the attitude error $\boldsymbol{\phi}^n$ is as follows:

$$\dot{\boldsymbol{\phi}}^n = -\boldsymbol{\omega}_{in}^n \times \boldsymbol{\phi}^n + \delta\boldsymbol{\omega}_{in}^n - \boldsymbol{C}_b^n \delta\boldsymbol{\omega}_{ib}^b \tag{4}$$

The velocity error differential equation is derived:

$$\delta\dot{\mathrm{v}}_{eb}^n = \left(\mathrm{C}_b^n \mathrm{f}_{ib}^b\right) \times \phi^n + \mathrm{C}_b^n \delta \mathrm{f}_{ib}^b - \left(2\omega_{ie}^n + \omega_{en}^n\right) \times \delta\mathrm{v}_{eb}^n - \left(2\delta\omega_{ie}^n + \delta\omega_{en}^n\right) \times \mathrm{v}_{eb}^n + \delta \mathrm{g}_{ib}^n \tag{5}$$

The corresponding position error differential equation is:

$$\begin{cases} \delta\dot{L} = \frac{1}{R_M+h}\delta V_N - \frac{V_N}{(R_M+h)^2}\delta h \\ \delta\dot{\lambda} = \frac{\sec L}{R_N+h}\delta V_E + \frac{V_E \sec L \tan L}{R_N+h}\delta L - \frac{V_E \sec L}{(R_N+h)^2}\delta h \\ \delta\dot{h} = \delta V_U \end{cases} \tag{6}$$

$\delta L, \delta\lambda$ and δh are latitude error, longitude error, and altitude error respectively.

In addition, the errors of the gyro and accelerometer are defined as:

$$\begin{aligned} \delta\boldsymbol{\omega}_{ib}^b &= \boldsymbol{\varepsilon}^b + \boldsymbol{w}_g, \dot{\boldsymbol{\varepsilon}}^b = \mathbf{0}_{3\times1}, \boldsymbol{w}_g \sim N(0, \boldsymbol{Q}_g) \\ \delta\boldsymbol{f}^b &= \nabla^b + \boldsymbol{w}_a, \dot{\nabla}^b = \mathbf{0}_{3\times1}, \boldsymbol{w}_a \sim N(0, \boldsymbol{Q}_a) \end{aligned} \tag{7}$$

Where $\delta\boldsymbol{\omega}_{ib}^b$ and $\delta\boldsymbol{f}^b$ represent the error values of the gyroscope and accelerometer respectively. The gyroscope drift error is $\boldsymbol{\varepsilon}^n$ and the constant drift error of the accelerometer is ∇^n, $\boldsymbol{w}_g$ and $\boldsymbol{w}_a$ are the corresponding random drift error respectively. $\boldsymbol{Q}_g$ and $\boldsymbol{Q}_a$ represent the corresponding covariance matrices of the random noise of the gyroscope and accelerometer, respectively.

3. Observability analysis based on SINS/DVL integrated navigation system

In this section, a study is carried out for the filtering model of the combined SINS/DVL navigation and its observability under hovering conditions to verify whether the theoretically unobservable state variables have inconsistent estimation, which is important for the later study of the group affine.

3.1. System modeling and error state equations

The SINS/DVL integrated navigation system generally corrects the navigation error of SINS through a feedback mechanism by means of the EKF algorithm, thus effectively suppressing the dispersion of the SINS error. The model structure is illustrated in Fig. 1.

Based on the equations in Section 2, we can conclude that in the SO-EKF system model, the attitude is defined on the SO(3), while the velocity and position are defined in euclidean space. It is obvious that the velocity, position and attitude states are in different spaces.

The state error of the SINS/DVL integrated navigation system is defined in n as follows:

$$\boldsymbol{X}_{SO-EKF} = [\boldsymbol{\phi}^n \quad \delta\boldsymbol{v}^n \quad \delta\boldsymbol{p}^n \quad \boldsymbol{\varepsilon}^n \quad \nabla^n]^T \tag{8}$$

The attitude angle error $\boldsymbol{\phi}^n = [\boldsymbol{\phi}^E \quad \boldsymbol{\phi}^N \quad \boldsymbol{\phi}^U]$ is the pitch, roll and heading angles, respectively. The velocity error $\delta\boldsymbol{v}^n = [\delta\boldsymbol{v}^E \quad \delta\boldsymbol{v}^N \quad \delta\boldsymbol{v}^U]$, eastward velocity, northward and celestial velocity, respectively. The position error $\delta\boldsymbol{P} = [\delta L \quad \delta\lambda \quad \delta h]$ is the latitude error, longitude error, and altitude error.

The corresponding error state model is given by:

$$\dot{\boldsymbol{X}}_{SO-EKF} = \boldsymbol{F}_{SO-EKF}\boldsymbol{X}_{SO-EKF} + \boldsymbol{G}_{SO-EKF}\boldsymbol{W} \tag{9}$$

Among them, the state transition matrix is $\boldsymbol{F}_{SO-EKF}$,the inertial guidance error state vector is $\boldsymbol{X}_{SO-EKF}$, the noise driving matrix is $\boldsymbol{G}_{SO-EKF}$, the process noise vector is $\boldsymbol{W}$.

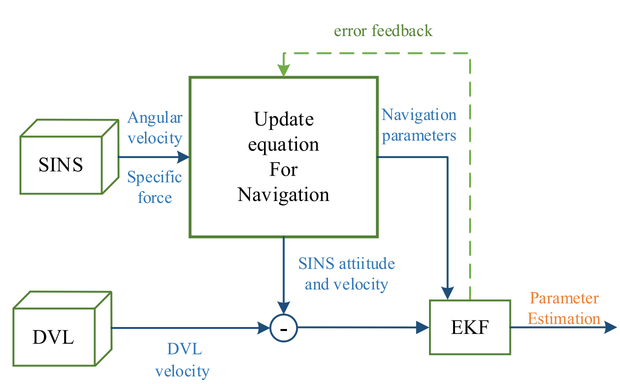

Fig. 1. The SINS/DVL integrated navigation model.

$$F_{SO-EKF}=\begin{bmatrix}-\omega_{in}^{n}\times & M_{12} & M_{13} & -C_b^n & 0_{3\times3}\\ C_b^n f^b\times & -(2\omega_{ie}^n+\omega_{en}^n) & M_{23} & 0_{3\times3} & C_b^n\\ 0_{3\times3} & M_{32} & M_{33} & 0_{3\times3} & 0_{3\times3}\\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3}\\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3}\end{bmatrix} \tag{10}$$

$$G_{SO-KF}=\begin{bmatrix}-C_b^n & 0_{3\times3}\\ 0_{3\times3} & C_b^n\\ 0_{9\times3} & 0_{9\times3}\end{bmatrix} \tag{11}$$

The parameter matrix $\boldsymbol{M}$ in the F_{SO-EKF} can be found in Appendix A.

In the SINS/DVL integrated navigation system, the velocities from both the SINS and DVL are typically selected as the measurement quantities. Initially, the SINS data and the DVL observation vector are provided in the local-level frame.

$$\widehat{\boldsymbol{v}}_{SINS}^{n}=\boldsymbol{v}_{SINS}^{n}+\delta\boldsymbol{v}_{SINS}^{n} \tag{12}$$

$$\widehat{\mathrm{v}}_{\mathrm{DVL}}^{\mathrm{n}}=\mathrm{C}_{\mathrm{n}}^{\mathrm{n}'}\mathrm{C}_{\mathrm{b}}^{\mathrm{n}}\mathrm{v}_{\mathrm{DVL}}^{\mathrm{b}}=[\mathrm{I}-(\phi^{\mathrm{n}}\times)]\mathrm{C}_{\mathrm{b}}^{\mathrm{n}}\mathrm{v}_{\mathrm{DVL}}^{\mathrm{b}} \tag{13}$$

Assuming that the calibration is completed, the influence of the installation error angle and the proportional factor are not considered for the time being. $\boldsymbol{\phi}^n\times$ is the antisymmetric matrix of the misalignment angle $\boldsymbol{\phi}^n$, $\widehat{\boldsymbol{v}}_{SINS}^{n}$ and $\widehat{\boldsymbol{v}}_{DVL}^{n}$ represent the calculated values of SINS and DVL in the n-frame. respectively. $\boldsymbol{v}_{DVL}^{b}$ is the DVL velocity in the b-frame.

By using Eq. (12) ~ (13), ignoring the quadratic error term, the measurement model can be calculated:

$$Z_{SINS/DVL}=\widehat{\boldsymbol{v}}_{SINS}^{n}-\widehat{\boldsymbol{v}}_{DVL}^{n}\approx-\left(\boldsymbol{v}_{DVL}^{n}\right)\times\boldsymbol{\phi}^{n}+\delta\boldsymbol{v}_{SINS}^{n} \tag{14}$$

Since the true values of SINS attitude and velocity cannot be obtained, the true values can be approximated by the measured values. The observation equation $\boldsymbol{Z}_{SINS/DVL}$ and measurement matrix $\boldsymbol{H}_{SINS/DVL}$ are as follows:

$$\boldsymbol{Z}_{SINS/DVL}=\boldsymbol{H}_{SINS/DVL}\boldsymbol{X}+\boldsymbol{V}_{SINS/DVL} \tag{15}$$

$$\boldsymbol{H}_{SO-EKF}=\begin{bmatrix}-\boldsymbol{v}_{DVL}^{n}\times & \boldsymbol{I}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\end{bmatrix} \tag{16}$$

In the traditional definition, the attitude angle error corresponding to $\boldsymbol{C}_b^n\widetilde{\boldsymbol{C}}_n^b$ in the form of euler angles is expressed as φ. The velocity and position errors are defined as the direct algebraic difference between the estimated and true values, assuming that φ is small. Thus, the feedback correction based on SO-EKF can be obtained

$$\begin{cases}\boldsymbol{C}_b^n=[\boldsymbol{I}_3+(\varphi\times)]\widetilde{\boldsymbol{C}}_b^n\\ \boldsymbol{v}^n=\widetilde{\boldsymbol{v}}^n-\delta\boldsymbol{v}^n\\ \boldsymbol{p}^n=\widetilde{\boldsymbol{p}}^n-\delta\boldsymbol{p}^n\end{cases} \tag{17}$$

3.2. Analytical method of observability

The AUV is required to perform fixed-point hovering tasks, which requires the AUV to maintain a quasi-stationary state and provide navigation information in a timely manner (Liu and Tan, 2024). At this time, the system cannot be dynamically stimulated or add other sensors to increase observability. Then it is necessary to understand the current observability and design filters to reduce inconsistency problems.

Therefore, this paper proposes an observability analysis method based on right null space analysis of PWCS combined with SVD. Different from the observability Gramian linear method used in reference (Itzik and Diamant, 2015), this section is aimed at the nonlinear SINS/DVL integrated navigation system. The PWCS extracted observability matrix replaces the total observable matrix to effectively reduce the complexity of analysis and calculation. Then, the SVD decomposition and the obtained singular values and corresponding singular vectors are used to establish the right null space for direct observation. The observability Gramian method needs to directly calculate the Gramian matrix of the time-varying system, which has a high computational complexity and is not directly quantifiable for the observability of a single state quantity. The PWCS combined with the SVD method based on the right null space can intuitively reflect the observability of the state quantity.

First, construct the observable matrix based on PWCS. For the j-th time period, it is defined as:

$$\begin{cases}\dot{\mathbf{X}}(t)=\mathbf{F}_j\mathbf{X}(t)+\mathbf{G}(t)\mathbf{W}(t)\\ \mathbf{Z}(t)=\mathbf{H}_j\mathbf{X}(t)+\mathbf{V}(t)\end{cases}\quad j=1,2,\dots,n \tag{18}$$

In the SINS/DVL integrated navigation system of the AUV, the $\boldsymbol{F}_{SO-EKF}$ and the $\boldsymbol{H}_{SO-EKF}$ are the state transfer matrix and the measurement matrix, respectively, as shown in Eq. (10) and Eq. (16). If the observability matrix is calculated at this time, the Grammmia matrix calculation is required, which is computationally cumbersome. When the time interval tj is small enough, the system can be approximately represented by PWCS, which can simplify the analysis process without affecting the system accuracy and system characteristics analysis. The total observability matrix (TOM) can be expressed as:

$$\boldsymbol{Q}_{(j)}=\begin{bmatrix}\boldsymbol{Q}_1\\ \boldsymbol{Q}_2e^{F_1\Delta_{t_{j-1}}}\\ \vdots\\ \boldsymbol{Q}_je^{F_j\Delta_{t_{j-1}}}\dots e^{F_1\Delta_{t_1}}\end{bmatrix} \tag{19}$$

Where: $\boldsymbol{F}_j$ is the state matrix constant of the system in the jth time period, Δ_{t_j} is the length of the jth time period, and $\boldsymbol{Q}_{(j)}$ is the TOM of the discrete PWCS.

$$\boldsymbol{Q}_j=\begin{bmatrix}H\\ HF\\ HF^2\\ \vdots\\ HF^{N-1}\end{bmatrix} \tag{20}$$

$$\boldsymbol{Q}_{s(j)}=\begin{bmatrix}\boldsymbol{Q}_1^T & \boldsymbol{Q}_2^T & \dots & \boldsymbol{Q}_j^T\end{bmatrix}^T \tag{21}$$

Perform singular value decomposition on $\boldsymbol{Q}_{s(j)}$ in the extracted observability matrix (XX):

$$\boldsymbol{Q}_{s(j)}=\boldsymbol{U}\sum\boldsymbol{V}^{\mathrm{T}} \tag{22}$$

Among them, $\boldsymbol{U}=[u_1\quad u_2\quad \dots\quad u_{rmn}]$ is $rmn\times rmn$ matrix, Σ is a singular value matrix, $\boldsymbol{\Sigma}=diag(\sigma_1,\sigma_2,\dots,\sigma_n)$ contains all singular values, and the singular values are arranged in descending order. $\sigma_1\geq\sigma_1\geq\dots\geq\sigma_n=0$, $\boldsymbol{V}^{\mathrm{T}}$ is a right singular vector matrix with dimension $n\times n$. Traditional methods use the singular value ratio (condition number: $\kappa=\sigma_{\max}/\sigma_{\min}$) to quantify the overall observability of the system rather than the observability of a single state variable. In addition, the singular value is only used as a quantitative indicator of the system as a whole, lacking the distinction between the contributions of state variables (Li et al., 2012).

This section proposes a method based on right null space analysis, which is different from the traditional method. Its purpose is to qualitatively analyze the unobservable state variables and directions in the system. The dimension of the null space and its basis vectors are used to characterize the unobservable directions, which reduces the computational complexity and clearly reflects which state variables of the system are completely unobservable, making it easier to compare the consistency with the actual observation results.

The last few columns in the right singular vector matrix $\boldsymbol{V}$ correspond to zero singular values. Specifically, for a zero singular value $\sigma_j=0$, its corresponding right singular vector v_j satisfies:

$$\boldsymbol{Q}_{vj} = 0 \tag{23}$$

The column vectors v_j corresponding to these zero singular values are the basis vectors of the right null space. If there are p zero singular values, then the corresponding right null space basis vector matrix $\boldsymbol{V}_{null}$ is:

$$\boldsymbol{V}_{null} = \boldsymbol{V}(:, n-p+1:n) \tag{24}$$

Among them, n is the number of columns of the observation matrix, p is the number of zero singular values. Each column of the matrix $\boldsymbol{V}_{null}$ is an unobservable direction.

Through the above method, we can obtain the unobservable state variables of the SINS/DVL integrated navigation system. Under stationary conditions, right null space analysis is performed.

The right null space $\boldsymbol{V}_{null_j} \in \mathbb{R}^{15\times3}$ of $\boldsymbol{Q}_j$ is

$$\boldsymbol{V}_{null_j} = \begin{bmatrix} \boldsymbol{V}_{null_j,\boldsymbol{\varepsilon}^E} & \boldsymbol{V}_{null_j,\nabla^N} & \boldsymbol{V}_{null_j,\nabla^E} \end{bmatrix} = \begin{bmatrix} 0_{3\times1} & 0_{3\times1} & 0_{3\times1} \\ 0_{3\times1} & 0_{3\times1} & 0_{3\times1} \\ 0_{3\times1} & 0_{3\times1} & 0_{3\times1} \\ b & 0 & 0 \\ 0_{3\times1} & 0_{3\times1} & 0_{3\times1} \\ 0 & c & 0 \\ 0 & 0 & d \end{bmatrix} \tag{25}$$

Where b, c, d are any non-zero integers. It can be concluded that when the AUV in a hovering condition, $\boldsymbol{\varepsilon}^E$, ∇^N, and ∇^E in the null space $\boldsymbol{V}_{null}$ of the observable matrix of the SINS/DVL integrated navigation system are unobservable.

4. SINS/DVL integrated navigation algorithm based on LSEGA-EKF

When the group model satisfies the group affine property, the updated state error model can achieve the property of being independent of the trajectory, so that the novel error state definition and the trajectory-independent state space model can avoid the problem of inconsistent state estimation (Herrmann and Kotyczka, 2024). Unfortunately, not all group state models satisfy the group affine property. As pointed out in the introduction, for high-precision inertial navigation applications, we should consider how to make the group state model satisfy the group affine property in the local-level frame.

4.1. Analysis of lie groups and lie algebras

Lie groups are groups with continuous properties. $SO(3)$ describes the rotation of a rigid body, while $SE(3)$ includes rotation and translation, and the representations are as follows:

$$SO(3) = \left\{ \mathbf{R} \in \mathbb{R}^{3\times3} \middle| \mathbf{R}\mathbf{R}^T = \mathbf{I}, \det(\mathbf{R}) = 1 \right\}. \tag{26}$$

$$SE(3) = \left\{ \mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \in \mathbb{R}^{4\times4} \middle| \mathbf{R} \in SO(3), \mathbf{t} \in \mathbb{R}^3 \right\} \tag{27}$$

Where $\mathbf{R}$ and $\mathbf{t}$ represent the rotation matrix and translation vector respectively, and the corresponding Lie algebras are:

$$\mathfrak{so}(3) = \left\{ \phi \in \mathbb{R}^3, \boldsymbol{\Phi} = \phi^\wedge \in \mathbb{R}^{3\times3} \right\} \tag{28}$$

$$\mathfrak{se}(3) = \left\{ \boldsymbol{\zeta} = \begin{bmatrix} \phi \\ \boldsymbol{\xi} \end{bmatrix} \in \mathbb{R}^6, \phi \in \mathbb{R}^3, \boldsymbol{\xi} \in \mathfrak{so}(3), (\boldsymbol{\zeta}\times) = \begin{bmatrix} (\phi\times) & \boldsymbol{\xi} \\ \mathbf{0}^T & 0 \end{bmatrix} \in \mathbb{R}^{4\times4} \right\} \tag{29}$$

$SE(3)$ and $SO(3)$ have the following mapping relationship:

$$\exp(\zeta\times) = \begin{bmatrix} \exp(\boldsymbol{\phi}\times) & \mathbf{J}\mathbf{v} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \tag{30}$$

where exp represents the exponential mapping of the matrix, and $\mathbf{J}$ can be approximated as the unit matrix.

4.2. Improve of group affine model of new SINS mechanization

The SINS update equations in the navigation frame are shown in Eqs. (2)–(4). Different from the traditional state definition method, this paper incorporates both attitude and velocity into the elements of the special euclidean group SE(3), that is,

$$\boldsymbol{\chi} = \begin{bmatrix} \mathbf{C}_b^n & \mathbf{v}^n \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \tag{31}$$

According to Lie group theory, the differential equation of $\boldsymbol{\chi}$ is expressed as $\dot{\boldsymbol{\chi}} = f_u(\boldsymbol{\chi})$. For the Lie group state of equation (11), the corresponding system model is:

$$f_u(\boldsymbol{\chi}) = \begin{bmatrix} \mathbf{C}_b^n(\boldsymbol{\omega}_{ib}^b\times) - (\boldsymbol{\omega}_{in}^n\times)\mathbf{C}_b^n & \mathbf{C}_b^n\mathbf{f}^b - (2\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n)\times\boldsymbol{v}_{eb}^n + \mathbf{g}_{ib}^n \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \tag{32}$$

$f_u(\boldsymbol{\chi})$ needs to satisfy group affinity to achieve the property that the state error is independent of the trajectory [40]:

$$f_u(\boldsymbol{\chi}_1\boldsymbol{\chi}_2) = f_u(\boldsymbol{\chi}_1)\boldsymbol{\chi}_2 + \boldsymbol{\chi}_1 f_u(\boldsymbol{\chi}_2) - \boldsymbol{\chi}_1 f_u(\mathbf{I}_{\mathrm{nd}})\boldsymbol{\chi}_2 \tag{33}$$

In the local-level frame, based on Eq. (31) and the SINS mechanical arrangement Eqs. (2)–(4) indicates that the state model does not satisfy Eq. (31) Consequently, model $f_u(\boldsymbol{\chi})$ is not affine and cannot ensure trajectory independence of the state error. Therefore, a transformation of the state model is necessary.

Prior to this, we conducted research on affine improvement methods in the field of large misalignment transfer alignment (Sheng et al., 2024), proving that the " log-linear " differential equation that satisfies group affine has a certain ability to solve the nonlinear problems caused by large misalignment. This paper does not consider " log-linear ", but conducts affine research in the field of SINS/DVL integrated navigation. In the local-level frame, the influence of the earth's rotation is considered, and not only the auxiliary velocity vector is introduced, but also the influence of the earth's gravity field is considered, and the gravity correction term is introduced. The purpose is to compensate for the gravity changes caused by the earth's rotation and changes in geographical location.

In the local-level frame, we introduce the auxiliary velocity vector and the gravity correction term to improve the affinity. First, we give the gravity correction term $\Delta\boldsymbol{g}_g$

$$\Delta\boldsymbol{g}_g = 2\gamma h/R \tag{34}$$

Where γ is the gravity gradient, h is the altitude (generally refers to the water depth in the field of underwater navigation), and R is the radius of the earth. For the SINS system, the gravity correction term is introduced into formula (3) for optimization:

$$\dot{\boldsymbol{v}}^n = \boldsymbol{C}_b^n\boldsymbol{f}^b - (\boldsymbol{\omega}_{in}^n\times)\boldsymbol{v}^n + \boldsymbol{g}^n + \Delta\boldsymbol{g}_g \tag{35}$$

Furthermore, the relationship between the gravity vector $\boldsymbol{g}^n$ and gravity $\overline{\boldsymbol{g}}^n$ is given as follows:

$$\boldsymbol{g}^n = \overline{\boldsymbol{g}}^n - (\boldsymbol{\omega}_{ie}^n\times)^2\boldsymbol{P}^n \tag{36}$$

At this time, the auxiliary velocity vector is introduced

$$\overline{\boldsymbol{v}}^n = \boldsymbol{v}^n + (\boldsymbol{\omega}_{ie}^n\times)\boldsymbol{P}^n \tag{37}$$

The velocity formula after introducing the gravity correction term and the auxiliary vector is:

$$\dot{\boldsymbol{v}}^n = \boldsymbol{C}_b^n \boldsymbol{f}^b - (\boldsymbol{\omega}_{in}^n \times)\overline{\boldsymbol{v}}^n + \overline{\boldsymbol{g}}^n + \Delta \boldsymbol{g}_g \tag{38}$$

In the local-level frame, after updating the SINS mechanical arrangement, $f_u(\boldsymbol{\chi})$ can be re-expressed as:

$$\begin{aligned} f_u(\boldsymbol{\chi}) &= \begin{bmatrix} \mathbf{C}_b^n(\boldsymbol{\omega}_{ib}^b\times) - (\boldsymbol{\omega}_{in}^n\times)\mathbf{C}_b^n & \mathbf{C}_b^n\boldsymbol{f}^b - (\boldsymbol{\omega}_{in}^n\times)\overline{\boldsymbol{v}}_{eb}^n + \overline{\mathbf{g}}^n + \Delta\boldsymbol{g}_g \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{C}_b^n & \mathbf{v}^n \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} (\boldsymbol{\omega}_{ib}^b\times) & \boldsymbol{f}_{ib}^b \\ \mathbf{0}_{1\times3} & 0 \end{bmatrix} + \begin{bmatrix} -(\boldsymbol{\omega}_{in}^n\times) & \overline{\mathbf{g}}_{ib}^n + \Delta\boldsymbol{g}_g \\ \mathbf{0}_{1\times3} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{C}_b^n & \mathbf{v}^n \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \\ &= \boldsymbol{\chi}\boldsymbol{W}_1 + \boldsymbol{W}_2\boldsymbol{\chi} \end{aligned} \tag{39}$$

According to Eq. (33), it is shown that in the local-level frame, the introduction of auxiliary velocity vectors and gravity correction terms to improve the dynamic equations can satisfy the group affine. Therefore, the error state model is independent of the global state estimation, and the introduction of the gravity term will not affect the group affine properties of the system. Because the verification environment of this paper is on land or water surface at the same longitude and latitude, the gravity correction term has little effect on the results of this paper, but it is a meaningful design for subsequent AUV seabed hovering or diving conditions.

A. Error model of LSEGA-EKF for SINS/DVL

For the SINS/DVL integrated navigation system, DVL can provide the velocity under the body frame as observation information. The choice of error model is relatively flexible. In order to save space, the comparison of error models will not be expanded for the time being. This section selects the left error model $\boldsymbol{\tau}_L = \widetilde{\boldsymbol{\chi}}^{-1}\boldsymbol{\chi}$ and carries out the subsequent related derivation work.

The left error state that satisfies group affine:

$$\boldsymbol{\tau}_L = \widetilde{\boldsymbol{\chi}}^{-1}\boldsymbol{\chi} = \begin{bmatrix} \widetilde{\boldsymbol{C}}_n^b\boldsymbol{C}_b^n & \widetilde{\boldsymbol{C}}_n^b(\boldsymbol{v}^n - \widetilde{\boldsymbol{v}}^n) \\ \boldsymbol{0}_{1\times3} & 1 \end{bmatrix} \triangleq \begin{bmatrix} \boldsymbol{\tau}_L^a & \boldsymbol{\tau}_L^v \\ \boldsymbol{0}_{1\times3} & 1 \end{bmatrix} \in SE(3) \tag{40}$$

Where $\widetilde{\boldsymbol{C}}_n^b$ and $\widetilde{\boldsymbol{v}}$ are the true attitude matrix and velocity, $\boldsymbol{\tau}_L^a$ and $\boldsymbol{\tau}_L^v$ are the attitude error and velocity error in the newly defined left error model (Sheng et al., 2024). According to the relationship between Lie group and Lie algebra, we can get

$$\boldsymbol{\tau}_L^a = \widetilde{\boldsymbol{C}}_n^b\boldsymbol{C}_b^n = \exp(\phi_L\times) \approx \boldsymbol{I}_{3\times3} + \boldsymbol{\phi}_L\times \tag{41}$$

$$\boldsymbol{\tau}_L^v = \widetilde{\boldsymbol{C}}_n^b(\boldsymbol{v}^n - \widetilde{\boldsymbol{v}}^n) = -\widetilde{\boldsymbol{C}}_n^b\delta\boldsymbol{v}^n \tag{42}$$

According to the above equation, the redefined error not only takes into account the size of the state vector, but also the direction difference. Different from the traditional error definition, the SE(3) matrix avoids the problem of spatial inconsistency. Moreover, this section transforms the equations so that the state space model realizes the group affine property, so that the state error is independent of the trajectory, overcoming the problem of inconsistent observability estimates that is prone to occur in SO-EKF.

In this section, based on the SINS mechanization that satisfies group affine, the new SINS error differential equation under the left error model is re-derived to form the SE(3) Lie group matrix, and the second-order small quantities are ignored. Finally, the attitude error equation is obtained as follows:

$$\dot{\boldsymbol{\phi}}_L = \delta\boldsymbol{\omega}_{in}^b\times + (\boldsymbol{\phi}\times\boldsymbol{\omega}_{ib}^b)\times - \delta\boldsymbol{\omega}_{ib}^b\times \tag{43}$$

The specific derivation process is shown in Appendix B. Similarly, the error model is redefined and a new velocity error differential equation is derived:

$$\begin{aligned} \dot{\boldsymbol{\tau}}_L^v &= \dot{\widetilde{\boldsymbol{C}}}_n^b(\delta\overline{\boldsymbol{v}}_{eb}^n) + \widetilde{\boldsymbol{C}}_n^b(\delta\dot{\overline{\boldsymbol{v}}}_{eb}^n) \\ &\approx -\widetilde{\boldsymbol{C}}_n^b(\overline{\boldsymbol{v}}_{eb}^n\times)\delta\boldsymbol{\omega}_{in}^n - \widetilde{\boldsymbol{\omega}}_{ib}^b\times\boldsymbol{\tau}_L^v - \widetilde{\boldsymbol{f}}_{ib}^b\times\boldsymbol{\phi}_L - \delta\boldsymbol{f}_{ib}^b \end{aligned} \tag{44}$$

The state equation of the LSEGA-EKF is defined as:

$$\dot{\boldsymbol{X}}_{LSEGA-EKF} = \boldsymbol{F}_{LSEGA-EKF}\boldsymbol{X}_{LSEGA-EKF} + \boldsymbol{G}_{LSEGA-EKF}\boldsymbol{W} \tag{45}$$

Thus, the state error vector is

$$\boldsymbol{X}_{LSEGA-EKF} = \begin{bmatrix} \boldsymbol{\phi}_L & \boldsymbol{\tau}_L^v & \delta\boldsymbol{p}_L & \delta\nabla & \delta\boldsymbol{\varepsilon} \end{bmatrix}^{\mathrm{T}} \tag{46}$$

Where $\boldsymbol{F}_{LSE-EKF}$ and $\boldsymbol{G}_{RSE-KF}$ are:

$$\boldsymbol{F}_{LSEGA-EKF} = \begin{bmatrix} -\widetilde{\boldsymbol{\omega}}_{ib}^b\times & 0_{3\times3} & 0_{3\times3} & -\boldsymbol{I} & 0_{3\times3} \\ -\widetilde{\boldsymbol{f}}_{ib}^b & -\widetilde{\boldsymbol{\omega}}_{ib}^b\times & 0_{3\times3} & 0_{3\times3} & -\boldsymbol{I} \\ 0_{3\times3} & \boldsymbol{M}_{23} & \boldsymbol{M}_{33} & 0_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \end{bmatrix} \tag{47}$$

$$\boldsymbol{G}_{LSEGA-EKF} = \begin{bmatrix} -\boldsymbol{I}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & -\boldsymbol{I}_{3\times3} \\ \boldsymbol{0}_{9\times3} & \boldsymbol{0}_{9\times3} \end{bmatrix} \tag{48}$$

First, we compare the formula derivation results of SO-EKF and LSEGA-EKF. The most significant difference in the state transfer matrices, as presented in Eqs. (10) and (45), is that the error state model of the LSEGA-EKF adheres to the group affine model, which ensures error independence and trajectory independence. Since there is no element of group state in $\boldsymbol{F}_{LSEGA-EKF}$, it is obvious that the nonlinear error $\boldsymbol{\tau}_r$ of LSEGA-EKF can be derived through $\boldsymbol{\tau}_L^a$ and $\boldsymbol{\tau}_L^v$. That is to say, as theorem (1) in the reference [40], the error state model is independent of the state estimate. In SO-EKF, it can be clearly seen that the traditional error state model is correlated, which confirms that the traditional error model is not state-independent.

When SINS/DVL integrated navigation system under poor observation condition, the evolution of errors can impact the system's true observability. However, the LSEGA-EKF, which is based on nonlinear error dynamics, possesses the property of independent propagation of state trajectories in the SINS/DVL. This ensures that the evolution of errors is not entirely dependent on the states.

B. The new Measurement model of SINS/DVL for LSEGA-EKF

In this section, based on the model definition of the left error, the SINS/DVL measurement equation based on LSEGA-EKF is derived, and a new feedback correction method is given for the corresponding error model.

First, for the left error model, the SINS/DVL error measurement equation of LSEGA-EKF is defined as:

$$\boldsymbol{Z}_{LSEGA}^v = (\boldsymbol{v}^n\times)\boldsymbol{C}_b^n\boldsymbol{\phi}_L - \widetilde{\boldsymbol{C}}_b^n\boldsymbol{\tau}_L^v \tag{49}$$

The specific derivation process of $\boldsymbol{Z}_{LSEGA}^v$ is shown in Appendix C. The observation equation and measurement matrix H are shown below

$$\boldsymbol{H}_{LSEGA,SINS/DVL} = \begin{bmatrix} (\boldsymbol{V}^n\times)\boldsymbol{C}_b^n & -\widetilde{\boldsymbol{C}}_b^n & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} \end{bmatrix} \tag{50}$$

In the above equation, $\boldsymbol{Z}_{LSE,\mathrm{INS/DVL}}^v$ is the measured value of SINS/DVL, which is the velocity difference provided by SINS and DVL.

In summary, the SINS/DVL integrated navigation algorithms for both the traditional SO-EKF and LSEGA-EKF have been completely derived. It can be clearly seen that the first difference is the state quantity, and the error differential equation changes accordingly. Therefore, in the derivation process, the LSEGA-EKF has a completely new state prediction, measurement equation and feedback correction.

5. Simulation and field experiments

This section focuses on the SINS/DVL integrated navigation system as the research object. Building on the observable analysis results from Section 3, the study investigates the inconsistent estimation impact of theoretically unobservable states through simulation. To ensure a comprehensive comparison, the simulation and measured experiments include not only the filtering results of LSEGA-EKF and SO-EKF, but also introduce the Lie group-based EKF (LSE-EKF) without group affine improvement. This comparison aims to highlight the advantages of the proposed method in this paper, particularly in terms of estimation consistency and performance in the presence of unobservable quantities.

5.1. Simulation test

According to the previous analysis, simulation experiments are carried out to test the effects of SO-EKF, LSEGA-EKF on SINS/DVL integrated navigation. To better compare the performance of different filters, the 100 Monte Carlo simulations were conducted. The simulation parameters are set as follows, the simulation time is 1000s and the AUV is assumed to remain stationary. The initial misalignment angle is set to $[0.2^{\circ} \quad 0.3^{\circ} \quad 1.2^{\circ}]$, the SINS sampling frequency is 100Hz, and the related parameters are as follows: the constant value drift of the gyroscope is $0.02^{\circ}/h$, and the random error is $0.0005^{\circ}/\sqrt{h}$; the constant value bias of the accelerometer is $100\mu g$, and the random error is $25\mu g/\sqrt{Hz}$.

Based on the observable analysis results in Section 3, the variance estimates of theoretically unobservable $\boldsymbol{\varepsilon}^E$, ∇^E and ∇^N are presented in Fig. 2.

For AUV in hovering condition, according to SINS/DVL observability analysis, it is known that $\boldsymbol{\varepsilon}^E$, ∇^E and ∇^N are unobservable states, so their variance should not change, because the change of covariance indicates that observability is improving. In the above figure, the variance of $\boldsymbol{\varepsilon}^E$, ∇^E and ∇^N is unexpectedly reduced, showing false observability. The above analysis can fully explain that SO-EKF has inconsistent state estimation problems. Inconsistent state estimation will make incorrect corrections in this direction, which may cause filter divergence.

Next, in order to verify the impact of inconsistency under hovering conditions, we compared the standard deviations and added LSEGA-EKF to the control group in the simulation. The simulation conditions assumed that the AUV remained hovering underwater, and the duration was set to 3600 to observe the integrated navigation process in a quasi-static environment. Fig. 3 is a comparison of the standard deviations of $\boldsymbol{\varepsilon}^E$, ∇^E and ∇^N.

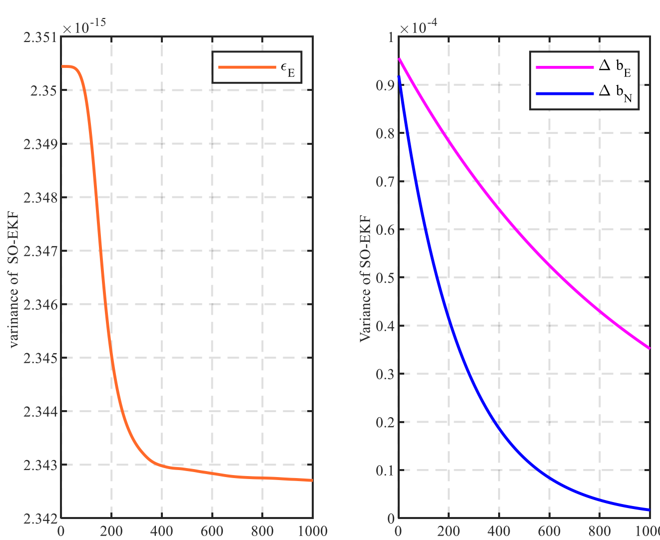

Fig. 2. State variance of SO-KF under static condition.

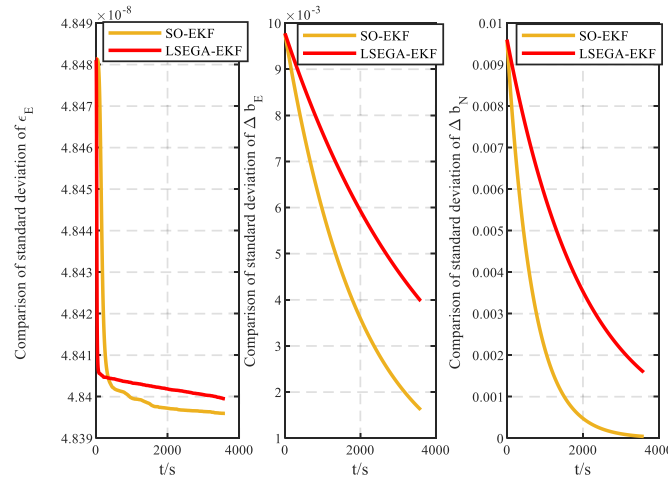

Fig. 3. Comparison of standard deviation of unobservable states.

From the above figure, the comparison of the standard deviations of different filter estimation errors shows that SO-EKF obviously has inconsistent state estimation, while the consistency of LSEGA-EKF is better than SO-EKF, ensuring the accuracy and stability of the SINS/DVL integrated navigation results under quasi-stationary poor-observation conditions.

Next, the convergence trends of heading angle estimation and east velocity error are used as observation objects affected by filtered inconsistency, because according to the Eq (Lv et al., 2020).- (He et al., 2023), the eastward gyroscope drift and the horizontal accelerometer bias are the main factors affecting the convergence of heading angle estimation. At the same time, the heading angle estimation will affect the eastward velocity calculation. The position state will hardly change in the quasi-stationary state, so it is not included in the simulation comparison.

In the SINS/DVL integrated navigation system under quasi-stationary conditions, considering that the position change during the simulation is essentially small and does not change significantly over time, we do not consider the position error comparison in the simulation, but include a detailed position error comparison in the final AUV experiment section. The simulation compares the heading angle and eastward velocity error estimates estimated by different filters, which can most intuitively observe how the originally unobservable state variables affect the filter estimates due to inconsistency problems. In addition, in order to highlight the advantages of affine transformation,

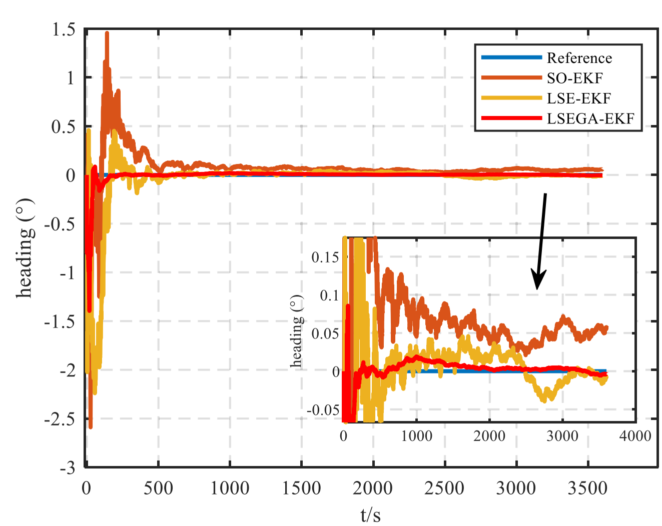

Fig. 4. Integrated navigation heading convergence.

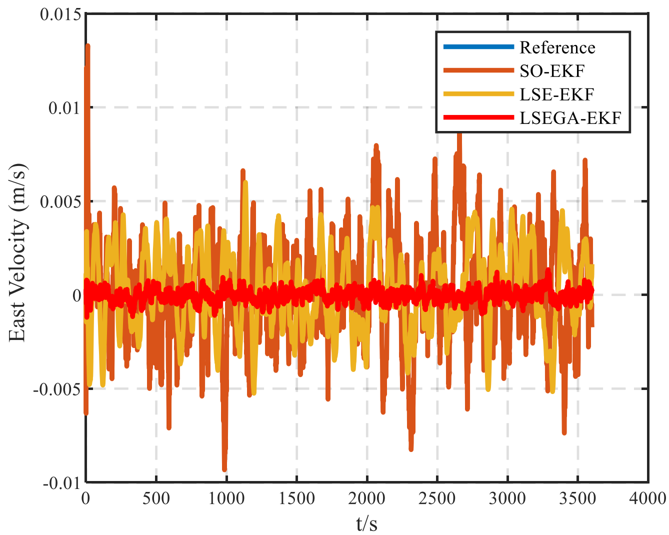

Fig. 5. Integrated navigation east velocity convergence.

LSE-EKF without affine transformation is added to the filtering results to provide a more comprehensive evaluation.

From Fig. 4 and Fig. 5 can clearly reflect that the SINS/DVL system is affected by the inconsistency problem under the poor observation conditions of the hovering condition. First, the heading error output by the traditional SO-EKF gradually diverges due to the inconsistency of the unobservables in the filtering process, and the eastward velocity error estimate fluctuates greatly. In addition, after the affine transformation mentioned above, the LSEGA-EKF is obviously better than the LSE-EKF in terms of the inconsistency problem of state estimation. The heading angle of the LSEGA-EKF can converge to an error of less than 0.02° from the reference value, and the velocity error estimate also fluctuates around the true value. The difference can be seen more intuitively from Tables I and II. The RSME value and Mean value under the LSEGA-EKF model are the smallest.The LSE-EKF model has a smaller estimation error and better variance preservation performance for unobservable states.

Based on the above simulation and error statistics, when the AUV is in hovering condition, some known unobservable state quantities will be directly misestimated due to inconsistent state estimation during the filtering process, which will directly lead to gradual deterioration or even divergence of the estimation. It can be concluded that under the condition of incomplete observation, the LSEGA-EKF model has better consistency of state estimation and more accurate navigation results. At the same time, it also proves the necessity of affine improvement.Table 3

5.2. Field experiments

In order to further verify the performance of the proposed method based on simulation, we then compared the integrated navigation performance and state consistency performance of different models by designing different vehicle experiments and lake test experiments.

(1) Land vehicle filed test

First, referring to the vehicle experiment process in AUV research in (Wang et al., 2019) – (Lyu et al., 2020), considering that the preparation process and implementation of AUV underwater experiments are difficult and costly, we first conducted a ground vehicle experiment to test the algorithm and verify the simulation results. Before that, it should be emphasized that there are differences in the dynamic characteristics between land vehicles and AUVs, but the vehicle is in a quasi-stationary motion state when starting in place and the AUV is in a hovering state, and the dynamic characteristics have certain similarities. In order to verify the versatility and effectiveness of the algorithm in different scenarios, the vehicle-mounted test can lay the foundation for subsequent AUV experiments.

The sensors in the experimental process only involve IMU and GNSS, and the positioning of GNSS is utilized to examine the positioning accuracy of the algorithm and to output the velocity measurement information instead of DVL. The parameters of IMU instrumentation during the experiment are shown in the following table, as shown in Table 3, and position reference information is provided by GNSS, and its performance index is shown in Table IV.

The equipment for the vehicle test is shown in Fig. 6. In order to construct an environment that is poor observation condition, the vehicle-mounted equipment is used as the research object, and the car is started, but no trajectory changes are generated. On this basis, the performance of SO-EKF,LSE-EKF and LSEGA-EKF is tested.

Taking the heading angle estimation as an example, the test is carried out according to the simulation setting parameters, the quasi-stationary time relative to SINS is 2600s, and the integrated navigation observation update time interval is 1s. The test results are shown in Fig. 7 and the error Table 4 is shown below.

We observe Fig. 6 and Table 3. In the vehicle-mounted integrated navigation test under the quasi-static environment, the performance under the poor observation condition is consistent with the simulation test. The accuracy of the heading angle estimation of SO-EKF is still

Table 1
Mean and RMS of the attitude error results.

Items	Index	RMSE	Mean
SO-KF	Heading error	0.0031	−8.48e-04
LSE-EKF		0.0023	−6.13e-04
LSEGA-EKF		0.0015	−1.12e-04

Table 2
Mean and RMS of the east velocity error results.

Items	Index	RMSE	Mean
SO-KF	East velocity error	0.0027	−2.94e-04
LSE-EKF		0.0021	−2.31e-04
LSEGA-EKF		0.0004	−9.61e-05

Table 3
Main parameters of IMU.

Items	Index	Values
Gyro	Bias	<0.02 *deg/h*
	Angular Random Walk	<0.01 deg/h
Accelerometer	Bias	<50 μg
	Random Walk Noise	<50 $\mu g/\sqrt{Hz}$

Fig. 6. Vehicle-mounted test equipment.

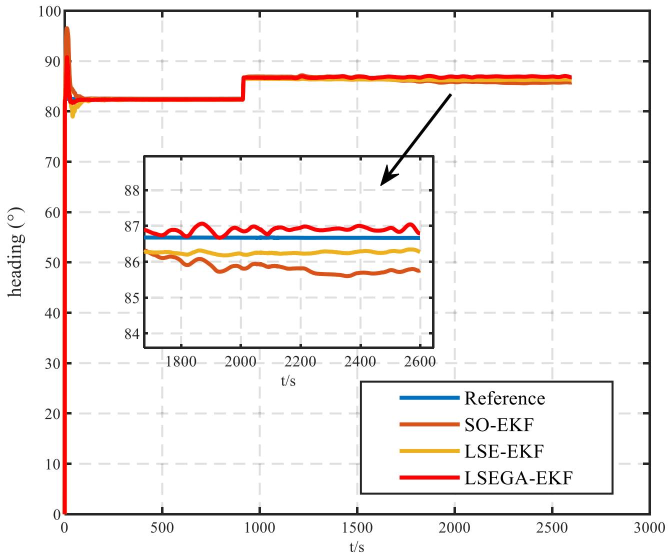

Fig. 7. Integrated navigation heading convergence result.

Table 4
Main parameters of IMU.

Model	accuracy	
	Horizontal	height
Single Point	1.5 m	2.5 m
RTK	1 cm + 1 ppm	1.5 cm + 1 ppm

Table 5
Mean and RMS of the Headind error results.

Items	Index	RMSE	Mean
SO-KF	Heading error	0.0210	−0.0022
LSE-EKF		0.0121	−0.0015
LSEGA-EKF		0.0078	−0.0013

lower than that of LSEGA-EKF. As shown in Fig. 6, after 2500s, the heading angle estimation of SO-EKF differs from the reference value by more than 1°, while that of LSEGA-EKF converges to an error within 0.5°. At the same time, the error values in Table 5 also illustrate the above conclusion, whether it is RMSE or Mean value, LSEGA-EKF performs better than LSE-EKF.

(2) Lake field test

In this section, in order to further deepen the vehicle experiment and verify the reliability of the algorithm, we decided to conduct a lake test experiment in a real underwater environment. A real AUV was used in the lake test experiment. The model was the ELF-P324, which has a carrying capacity of more than 20 KG, a heading control accuracy of ±0/5°, and a depth holding accuracy of ±0.1m. The AUV adopts modular design technology, and the positions of the compartments can be arbitrarily exchanged, which is convenient for expanding new functional compartments. This experiment is equipped with high-precision SINS and DVL and other core modules, which can complete underwater environment detection, underwater search and positioning, inspection and other tasks. The Elf-P324 is shown in Fig. 8.

We selected Shazhou Lake of Zhangjiagang city as the experimental site due to its favorable environment and stable water conditions. The absence of turbulent water flow minimizes the risk of significant displacement or angular deviations of the AUV, ensuring that no positive incentives artificially enhance the observability of the integrated

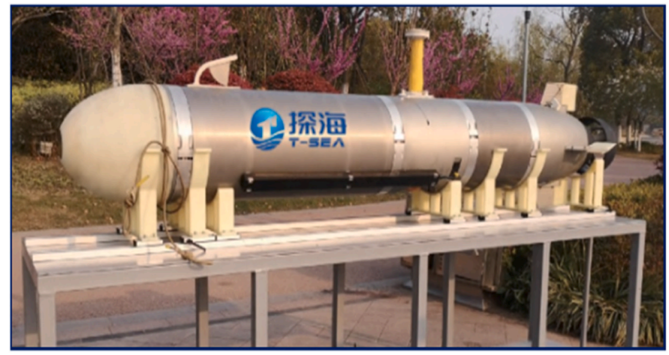

Fig. 8. Elf-P324 AUV Physical equipment.

navigation system. This choice aligns with our requirement for a partially observable experimental environment.

In the initial stage of the experiment, the AUV was placed on the water surface, fixed close to the dock, and kept stationary to ensure that the navigation system could accurately calibrate and initialize its sensor data, as shown in Fig. 9.

After ensuring that all sensors and equipment are working properly, real-time sensor data collection begins. The INS parameters are consistent with those in Table 3. The DVL performance parameters are shown in Table 6.

The SINS and DVL systems are turned on and connected to the central control system of the AUV (see Table 7). After initial alignment, the data collected in the lake test are input into the SINS/DVL integrated navigation algorithm. We selected 1200s of experimental time. During the interception time, the AUV was in a quasi-stationary state. The DVL velocity output is shown in Fig. 10. The velocity basically fluctuates slightly around 0. This shows that the AUV is in poor observation conditions and there is no incentive to improve its observability. We need to verify the adaptability and accuracy of the proposed algorithm in underwater environments under this condition.

The heading angel estimation in Fig. 11 are consistent with the previous vehicle-mounted results. There is a large deviation between the heading angle estimation of SO-EKF and the reference value. LSEGA-EKF can estimate the heading angle more accurately and has a smaller estimation error than LSE-EKF. Similarly, the RMSE and Mean values in the error Table 6 are consistent with the error diagram. We can clearly see from the statistical results that the use of the LSEGA-EKF has better error estimation effect.

From the perspective of AUV engineering application, the

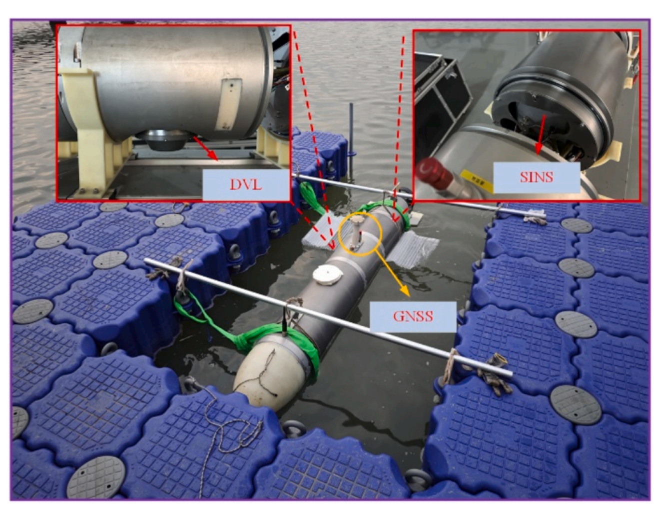

Fig. 9. Equipment of AUV-mounted field test.

Table 6
Main parameters of DVL.

ITEMS	Bottom-Tracking	Water-Tracking
Measuring range	0.2m–89m	1.9m–47m
Velocity measurement range	±9 m/s	±12 m/s
Sampling frequency	Adaptive frequency,max 12Hz	
Max working depth	300m	

Table 7
Mean and RMS of the error from AUV test.

Items	Index	RMSE	Mean
SO-KF	Heading angel	0.5246	−0.4945
LSE-EKF		0.5011	−0.4890
LSEGA-EKF		0.1142	−0.1035

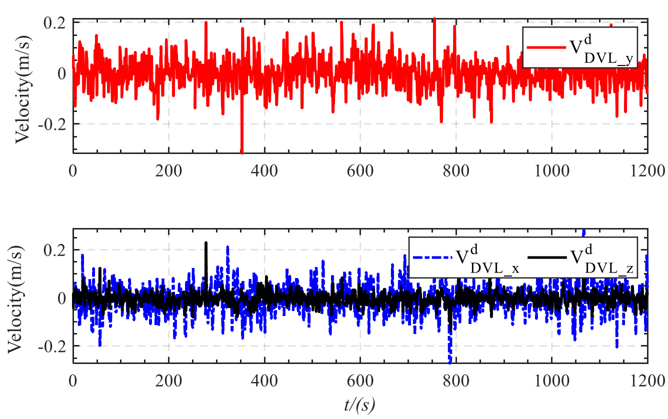

Fig. 10. The DVL velocity output.

positioning error estimation of the method proposed in this paper is verified. We compare the SINS/DVL navigation positioning errors of pure SINS solution, SO-EKF, LSE-EKF and LSEGA-EKF, as shown in Fig. 12. It can be clearly seen that the position error of pure SINS solution reaches more than 800m. Although the error can be greatly reduced by SO-EKF, the position error based on LSEGA-EKF is significantly smaller than that of SO-EKF, and the divergence trend of LSE-EKF is reduced. Therefore, under real hovering conditions, the LSEGA-EKF method can not only reduce the heading angle estimation error, but also play a significant role in improving the positioning accuracy of the SINS/DVL integrated navigation system.

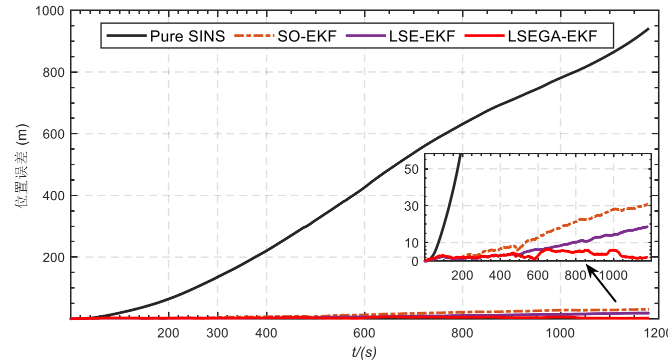

Fig. 12. AUV Positioning error of different algorithms.

In this SINS/DVL experiment, we verified the causes and impacts of the inconsistency in state estimation explained in the previous article, and compared the performance of the traditional method with the SINS/DVL method proposed in this paper under poor observation conditions.

6. Conclusions

Aiming at the poor-observation condition of AUV in hovering condition, this paper proposes a method based on right null space PWCS combined with SVD, which explains the inconsistency problem of SO-EKF on SINS/DVL integrated navigation system. The theory of Lie group and Lie algebra is introduced, and the nonlinear state error and measurement model of the system are redefined. In order to avoid the limitations of the commonly used inertial frame, auxiliary vectors and gravity correction terms are introduced in the local-level frame, and group affine is derived. A SINS/DVL integrated navigation method based on LSEGA-EKF is proposed. Finally, the proposed method is compared with different methods through simulation experiments and field tests. The results show that it not only effectively reduces the inconsistency caused by unobservable state variables, but also enhances the stability of the filtering system in hovering condition.

CRediT authorship contribution statement

Guangrun Sheng: Writing – original draft, Methodology. **Xixiang Liu:** Writing – review & editing, Funding acquisition. **Yingxue Zhang:**

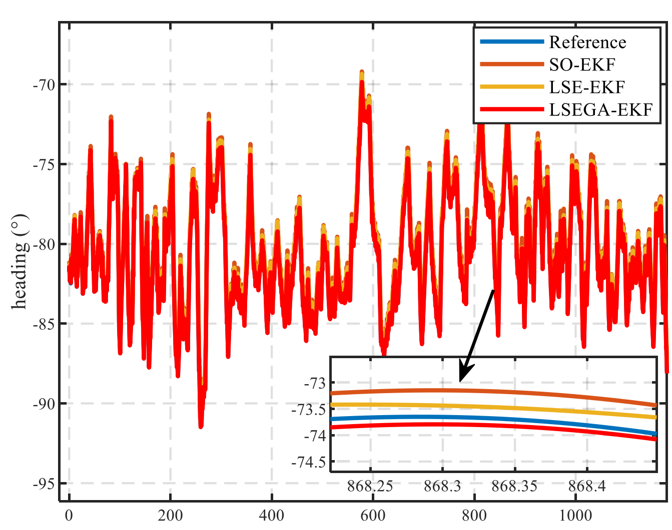

Fig. 11. AUV heading angel estimation.

Investigation. **Qiantong Shao:** Software. **Hao Xu:** Software. **Xiangzhi Cheng:** Validation. **Xiaoqiao Yuan:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A. The parameter matrix $\boldsymbol{M}$ in the $\boldsymbol{F}_{SO-EKF}$

$$\begin{cases}
\boldsymbol{M}_{12} = \begin{bmatrix} 0 & -\dfrac{1}{R_M+h} & 0 \\ \dfrac{1}{R_N+h} & 0 & 0 \\ \dfrac{\tan L}{R_N+h} & 0 & 0 \end{bmatrix}, \boldsymbol{M}_{13} = \begin{bmatrix} 0 & 0 & \dfrac{V_N}{(R_M+h)^2} \\ -\omega_{ie}\sin L & 0 & -\dfrac{V_E}{(R_N+h)^2} \\ \omega_{ie}\cos L + \dfrac{V_E}{(R_N+h)\cos^2 L} & 0 & -\dfrac{V_E\tan L}{(R_N+h)^2} \end{bmatrix} \\
\boldsymbol{M}_{23} = \begin{bmatrix} 2\omega_{ie}(V_N\cos L + V_U\sin L) + \dfrac{V_E V_N}{(R_N+h)\cos^2 L} & 0 & \dfrac{V_E V_U - V_E V_N\tan L}{(R_N+h)^2} \\ -2V_E\omega_{ie}\cos L - \dfrac{V_E^2}{(R_N+h)\cos^2 L} & 0 & \dfrac{V_N V_U}{(R_M+h)^2} + \dfrac{V_E^2\tan L}{(R_N+h)^2} \\ -2V_E\omega_{ie}\sin L & 0 & -\dfrac{V_N^2}{(R_M+h)^2} - \dfrac{V_E^2}{(R_N+h)^2} \end{bmatrix} \\
\boldsymbol{M}_{32} = \begin{bmatrix} 0 & \dfrac{1}{R_M+h} & 0 \\ \dfrac{1}{(R_N+h)\cos L} & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \boldsymbol{M}_{33} = \begin{bmatrix} 0 & 0 & -\dfrac{V_N}{(R_M+h)^2} \\ \dfrac{V_E\sin L}{(R_N+h)\cos^2 L} & 0 & -\dfrac{V_E}{(R_N+h)^2\cos L} \\ 0 & 0 & 0 \end{bmatrix}
\end{cases}$$

B. The derivation process of attitude error equation $\dot{\boldsymbol{\phi}}_L$

$$\begin{gathered}
\dot{\boldsymbol{\phi}}_L = \dot{\widetilde{\boldsymbol{C}}}{}_n^b \boldsymbol{C}_b^n + \widetilde{\boldsymbol{C}}_n^b \dot{\boldsymbol{C}}_b^n \\
= \left(\widetilde{\boldsymbol{C}}_n^b(\widetilde{\boldsymbol{\omega}}_{in}^n \times) - (\widetilde{\boldsymbol{\omega}}_{ib}^b \times)\widetilde{\boldsymbol{C}}_n^b\right)\boldsymbol{C}_b^n + \widetilde{\boldsymbol{C}}_n^b\left(\boldsymbol{C}_b^n(\boldsymbol{\omega}_{ib}^b \times) - (\boldsymbol{\omega}_{in}^n \times)\boldsymbol{C}_b^n\right) \\
= \widetilde{\boldsymbol{C}}_n^b(\widetilde{\boldsymbol{\omega}}_{in}^n \times)\boldsymbol{C}_b^n - (\widetilde{\boldsymbol{\omega}}_{ib}^b \times)\widetilde{\boldsymbol{C}}_n^b \boldsymbol{C}_b^n + \widetilde{\boldsymbol{C}}\boldsymbol{C}_n^b \boldsymbol{C}_b^n(\boldsymbol{\omega}_{ib}^b \times) - \widetilde{\boldsymbol{C}}_n^b(\boldsymbol{\omega}_{in}^n \times)\boldsymbol{C}_b^n \\
\approx \widetilde{\boldsymbol{C}}_n^b\left(\left(\widetilde{\boldsymbol{\omega}}_{in}^n - \boldsymbol{\omega}_{in}^n\right) \times\right)\boldsymbol{C}_b^n - (\widetilde{\boldsymbol{\omega}}_{ib}^b \times)(\boldsymbol{I}_{3\times3} + \boldsymbol{\phi} \times) + (\boldsymbol{I}_{3\times3} + \boldsymbol{\phi} \times)(\omega_{ib}^b \times) \\
= (\boldsymbol{I}_{3\times3} + \boldsymbol{\phi} \times)(\delta\boldsymbol{\omega}_{in}^b \times) - (\widetilde{\boldsymbol{\omega}}_{ib}^b \times)(\boldsymbol{\phi} \times) - \widetilde{\boldsymbol{\omega}}_{ib}^b \times + \boldsymbol{\omega}_{ib}^b \times + (\boldsymbol{\phi} \times)(\boldsymbol{\omega}_{ib}^b \times) \\
\approx = \delta\boldsymbol{\omega}_{in}^b \times + \left(\boldsymbol{\phi} \times \boldsymbol{\omega}_{ib}^b\right) \times - \delta\boldsymbol{\omega}_{ib}^b \times
\end{gathered}$$

C. The derivation process of measurement equation $\boldsymbol{Z}_{LSEGA}^v$

$$\begin{aligned}
\boldsymbol{Z}_{LSEGA}^v &= \delta\boldsymbol{v}^n \approx \widetilde{\boldsymbol{v}}^n - \left[\boldsymbol{I} - ({\boldsymbol{\phi}_L}^n \times)\boldsymbol{C}_b^n\right]\boldsymbol{v}^b \\
&= \widetilde{\boldsymbol{v}}^n + \boldsymbol{C}_b^n({\boldsymbol{\phi}_L}^n \times)\boldsymbol{v}^b - \boldsymbol{C}_b^n\boldsymbol{v}^b \\
&= -\boldsymbol{C}_b^n(\boldsymbol{v}^b \times)\boldsymbol{C}_n^b\boldsymbol{C}_b^n\boldsymbol{\phi}_L - \widetilde{\boldsymbol{C}}_b^n\boldsymbol{\tau}_L^v \\
&= (\boldsymbol{v}^n \times)\boldsymbol{C}_b^n\boldsymbol{\phi}_L - \widetilde{\boldsymbol{C}}_b^n\boldsymbol{\tau}_L^v
\end{aligned}$$

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